JOS - Josephson Junction

Signature Sheet

Student's Name _____ Partner's Name _____

Suggested reading to start with (see the end of the manual for more):

1. 111B JOS manual
2. Clarke, John, "**The Josephson Effect and e/h**," American Journal of Physics, Vol. 38, No. 9, September 1970, pp. 1071-1095.
3. Four Wire Measurement **Four Wire Measurement**
4. **JOS video**. Note in order to view this, you must be signed into your berkeley.edu Google account.
5. You must have completed the **EHS 109 Cryogen Safety training** prior to starting this lab, as you will be working with liquid nitrogen.

Pre-Lab Discussion Questions

It is your responsibility to discuss this lab with an instructor before your first day of your scheduled lab period. This signed sheet must be included as the first page of your report. Without it you will lose grade points. You should be prepared to discuss at least the following before you come to lab:

1. What is a Josephson Junction?
2. What is a Cooper Pair?
3. How does the DC Josephson effect work? What is the AC Josephson effect? How are they useful?
4. Why is it important to know the number 2e/h; what does it mean?
5. How do you construct the Josephson Junction used in this experiment? (Hint 4 wire junction)
6. Explain how you remove and install the tip into the junction assembly, and how you remove and install the junction assembly into the probe. Which way do you turn the set screws?
7. What are the three things most likely to greatly slow down your progress or legitimately break the system? (And so you must be careful with them)
8. Did you notify a professor or GSI about filling the liquid helium tank one day before starting the lab?
9. Get replacement set screw(s) for mounting the tip assembly in the probe from a GSI and replace them with the GSI. (This is to help prevent them from getting stripped.)

Staff Signature _____ Date _____

Completed before the first day of lab? (Circle one) Yes / No

Mid-Lab Discussion Questions

1. By day 4, show your DC effect to an instructor. What changes did you make in order to improve your observation?

Staff Signature _____ Date _____

Completed by day 4 of lab? (Circle one) Yes / No

Checkpoint Signatures

1. Low-Frequency Oscillator

   Staff Signature _____

2. Steps

   Staff Signature _____

3. AC Effect

   Staff Signature _____

4. Josephson Constant

   Staff Signature _____